📄 Digital Forensic II – Slip 1 Solution

✅ Q1: Extract metadata from an email header and analyze its origin and routing.

a) Extract metadata from email header

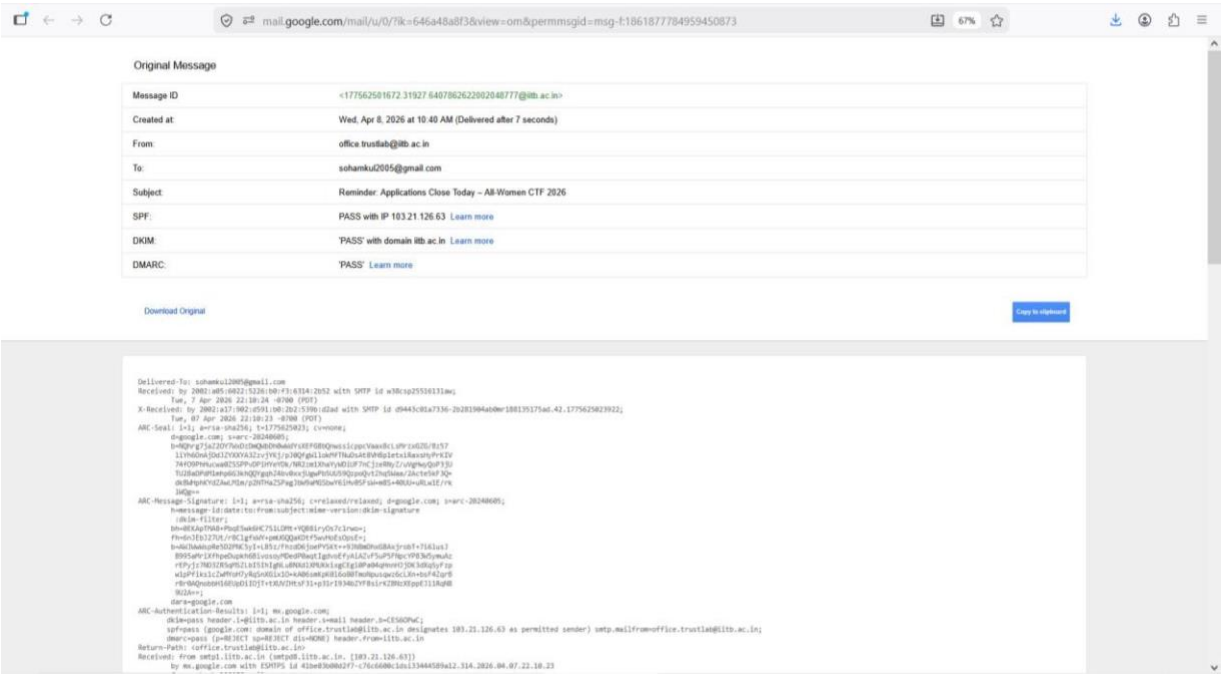
Steps:

1. Open the email (Gmail/Outlook)
2. Click **“Show Original” / “View Source”**
3. Observe header fields

Important fields:

- **From:** Sender’s email address
- **To:** Receiver’s email
- **Date:** Time of sending
- **Received:** Shows route of email servers
- **Message-ID:** Unique identifier
- **IP Address:** Origin of email

Conclusion:
Metadata helps trace the **origin and path** of the email.



- Email header page (Show Original)

b) Identify forged email headers and detect spoofing attempts.

Steps:

1. Compare **From** and **Return-Path**
2. Check **Received fields (bottom → top)**
3. Identify unknown/suspicious IP addresses 4. Verify:
   - **SPF = fail** ○ **DKIM = fail**

Indicators of spoofing:

- Fake domain (e.g., paytm-secure.xyz)
- Mismatch in sender details
- Authentication failure

Conclusion:

If mismatch or authentication fails → email is **spoofed**


```
Delivered-To: josh@techlicious.com
Received: by 2002:adf:f884:0:0:0:0:0 with SMTP id u4-v6csp3380720wrp;
        Wed, 31 Oct 2018 06:57:13 -0700 (PDT)
X-Google-Smtp-Source: AJdET5eTQ4uiy/jeMsfgpHbHMmTwAgqUTL+Cck05Zk2s7Izm8OYdMGd2RlXlNyLvW4JKjap0d43V
X-Received: by 2002:adf:dc87:: with SMTP id r7-v6mr3152022wrj.143.1540994233699;
        Wed, 31 Oct 2018 06:57:13 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1540994233; cv=none;
        d=google.com; s=arc-20160816;
        b=FNPAuRFdYt1Mcf0g3Vi6JwQSeNKarBb3+0thNT619PTxASnzG/baGBBMd0e7S4jGOC
         fFRN+KTOTO3tL4eVLkP9C/iuSFa9JhV2R5zZtaojNyH6cZ3fzjn3fmLGjFcwma0UK85h
         LWQIa9i/Q3S1+UkFR+1aXocMVd6C0B5J1XhMvAFlN1OLc6usT37Vpbtfn8g6oUxrP+tC
         O7i9JQRfpM3h/O5vWmcIIiI11UDxPiU/c/9GZZF5295mUGCM8Yy7YLcxF6qLN1Ep1sCX
         Ak6H/zik4BD90lGqhzW71C35QHnN2vELncMmXN0Q8WwiyV+RFeRT2tuVzvaW4J4Fo0re
         vmPg==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20160816;
        h=date:message-id:reply-to:errors-to:importance:from:subject:to;
        bh=9mqxVecZ4Hf+QFs63lC04Ks02gQdvvXnDATjlTaIegY=;
        b=Vb9hckd1oialslM959jUbxPFOli4+tDlBFSDZhMB1vZvWBOD3rQmMRiJAsK+z5HP8U
         uyIQNl1O3ShySoHQPfZPTwRZhofHo2j5tCLfncOYsBfUInNZNwHGkgt2tNkXUiHXf/5W
         JjB5MmTNv5/JENjRaAnwjNVz4USBhB07vu4JrsGxNUUHfjDQMA8iQRk2wg4bDCiOrRGW
         II69WtTPdB1ANhU19uFMoIRDVsinAlTFc/vm2ehTXj3/e8ViobT0NZyh3WYb69xsHTgO
         Se/r8tk6f63H+QwAg1IO1LgpB7jqGSyL1VfhNRM6AvNaVydqGN3YD2F9rInvgTh3vGXU
         qQXg==
ARC-Authentication-Results: i=1; mx.google.com;
       spf=softfail (google.com: domain of transitioning josh@techlicious.com does not designate
46.167.245.206 as permitted sender) smtp.mailfrom=josh@techlicious.com;
      dmarc=fail (p=NONE sp=NONE dis=NONE) header.from=techlicious.com
Return-Path: <josh@techlicious.com>
Received: from emkei.cz (emkei.cz. [46.167.245.206])
        by mx.google.com with ESMTPS id l5-v6si3316066wrt.171.2018.10.31.06.57.13
        for <josh@techlicious.com>
        (version=TLS1_2 cipher=ECDHE-RSA-AES128-GCM-SHA256 bits=128/128);
        Wed, 31 Oct 2018 06:57:13 -0700 (PDT)
Received-SPF: softfail (google.com: domain of transitioning josh@techlicious.com does not designate
46.167.245.206 as permitted sender) client-ip=46.167.245.206;
Authentication-Results: mx.google.com;
       spf=softfail (google.com: domain of transitioning josh@techlicious.com does not designate
46.167.245.206 as permitted sender) smtp.mailfrom=josh@techlicious.com;
      dmarc=fail (p=NONE sp=NONE dis=NONE) header.from=techlicious.com
Received: by emkei.cz (Postfix, from userid 33) id 4782ED6628; Wed, 31 Oct 2018 14:57:13 +0100 (CET)
To: josh@techlicious.com
Subject: This is a spoof email scam test
From: Josh Scammer <josh@techlicious.com>
X-Priority: 3 (Normal)
Importance: Normal
Errors-To: josh@techlicious.com
Reply-To: josh@techlicious.com
Content-Type: text/plain; charset=utf-8
Message-Id: <20181031135713.4782ED6628@emkei.cz>
Date: Wed, 31 Oct 2018 14:57:13 +0100 (CET)

This is a spoof email scam test
```


- Header showing SPF/DKIM fail (optional)

c) Investigate an email phishing attack using header analysis.

Steps:

1. Analyze sender email domain
2. Trace IP using header
3. Check authentication:
    - SPF
    - DKIM
4. Inspect suspicious links

Indicators of phishing:

- Fake domain names
- Urgent messages (e.g., “account blocked”)
- Suspicious links

Conclusion:
Header analysis helps detect **phishing attempts and attacker source**

✅ Q2: Describe the steps you followed to recover contacts using your Google account

Steps:

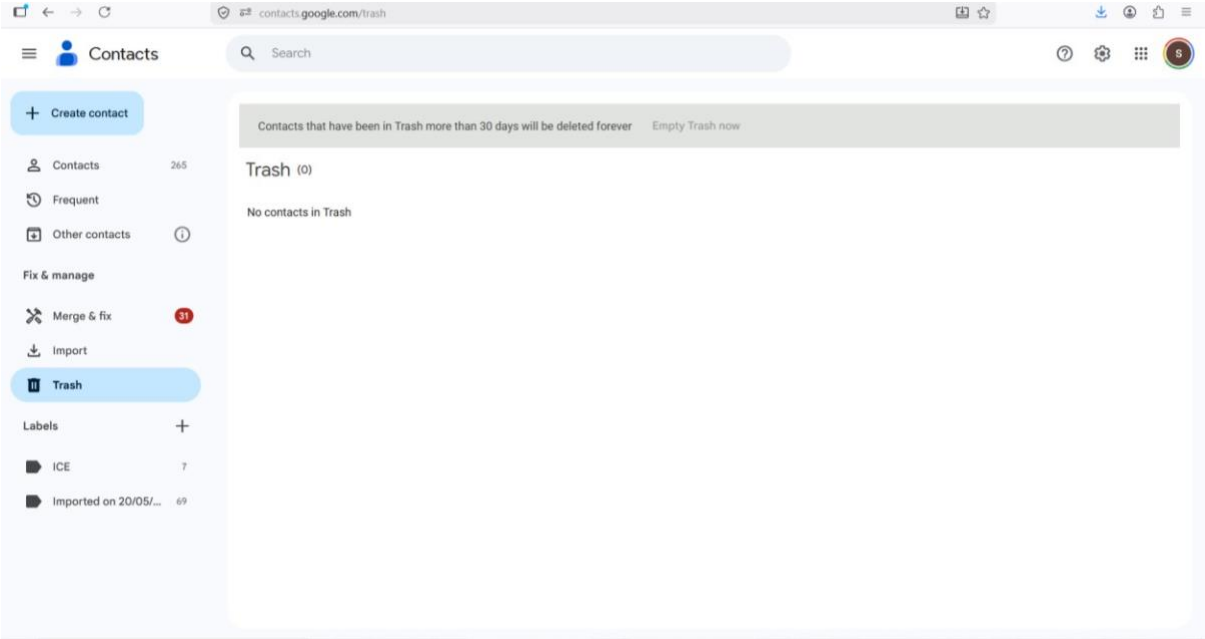
1. Go to **Google Contacts**
2. Login with your account
3. Click **Trash / Bin**
4. Select deleted contacts
5. Click **Restore**

Alternative method:

- Click **Settings → Undo Changes**
- Restore contacts (up to 30 days)

Forensic Points:

- Data source: Google Cloud
- Recovery type: Backup restore
- Time limit: ~30 days
- Requires authentication



- Google Contacts restore page

📄 Digital Forensic II – Slip 2 Solution

✅ Q1: Extracting device information of your Android device using ADB. Create a report of your findings.

Steps:

1. Enable **Developer Options** on phone
2. Turn ON **USB Debugging**
3. Connect phone to PC via USB 4. Open terminal / command prompt 5. Run commands:

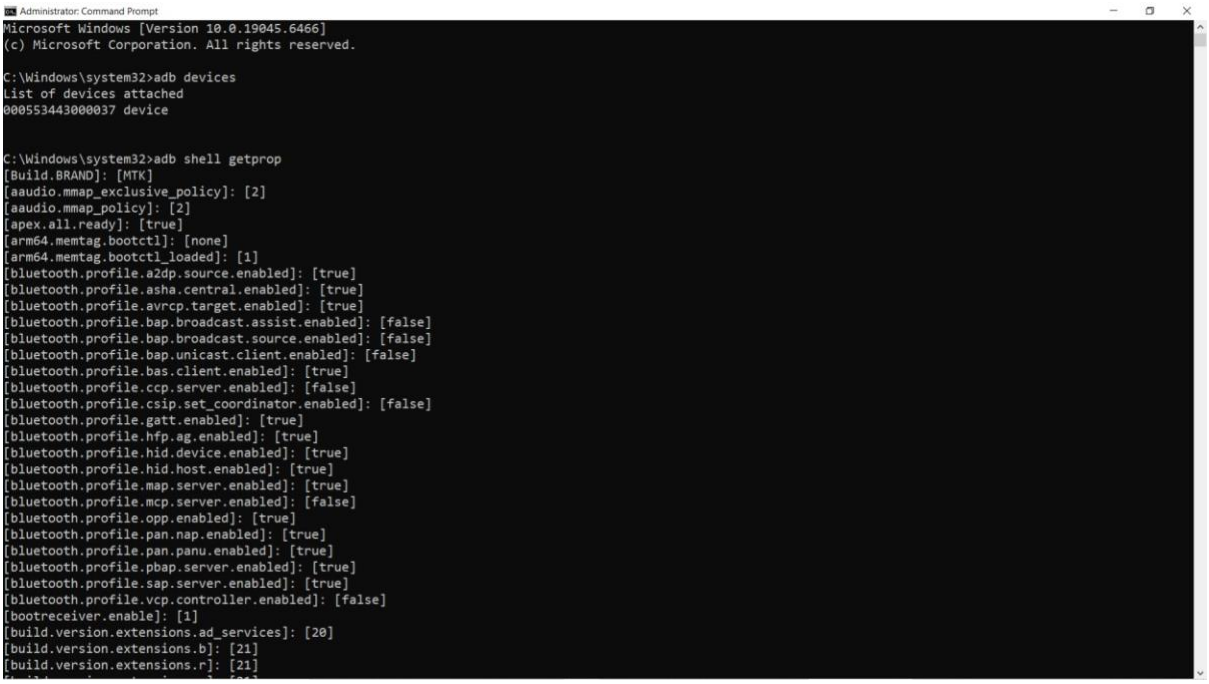
```
adb devices adb
shell getprop
```

Important Information Extracted:

- **Manufacturer:** `ro.product.manufacturer`
- **Model:** `ro.product.model`
- **Android Version:** `ro.build.version.release`
- **Device ID:** `adb devices`

Conclusion:

ADB helps retrieve **basic device info** useful for forensic analysis.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>adb devices
List of devices attached
000553443000037 device


C:\Windows\system32>adb shell getprop
[Build.BRAND]: [MTK]
[aaudio.mmap_exclusive_policy]: [2]
[aaudio.mmap_policy]: [2]
[apex.all.ready]: [true]
[arm64.memtag.bootctl]: [none]
[arm64.memtag.bootctl_loaded]: [1]
[bluetooth.profile.a2dp.source.enabled]: [true]
[bluetooth.profile.asha.central.enabled]: [true]
[bluetooth.profile.avrcp.target.enabled]: [true]
[bluetooth.profile.bap.broadcast.assist.enabled]: [false]
[bluetooth.profile.bap.broadcast.source.enabled]: [false]
[bluetooth.profile.bap.unicast.client.enabled]: [false]
[bluetooth.profile.bas.client.enabled]: [true]
[bluetooth.profile.ccp.server.enabled]: [false]
[bluetooth.profile.csip.set_coordinator.enabled]: [false]
[bluetooth.profile.gatt.enabled]: [true]
[bluetooth.profile.hfp.ag.enabled]: [true]
[bluetooth.profile.hid.device.enabled]: [true]
[bluetooth.profile.hid.host.enabled]: [true]
[bluetooth.profile.map.server.enabled]: [true]
[bluetooth.profile.mcp.server.enabled]: [false]
[bluetooth.profile.opp.enabled]: [true]
[bluetooth.profile.pan.nap.enabled]: [true]
[bluetooth.profile.pan.panu.enabled]: [true]
[bluetooth.profile.pbap.server.enabled]: [true]
[bluetooth.profile.sap.server.enabled]: [true]
[bluetooth.profile.vcp.controller.enabled]: [false]
[bootreceiver.enable]: [1]
[build.version.extensions.ad_services]: [20]
[build.version.extensions.b]: [21]
[build.version.extensions.r]: [21]
```

- Terminal showing `adb devices` and `getprop`

✅ Q2: Use WinHex to find the last access time of Google Drive on your computer?

Steps:

1. Open WinHex
2. Go to **Tools → Open Disk** 3. Load system disk
4. Navigate to:
5. `C:\Users\Admin\AppData\Local\Google\DriveFS`
6. Open relevant file (e.g., `cello_asset_history`)
7. View metadata in right panel
8. Note **Last Access Time**

Result:

- Last Access Time found: **02-04-2026 11:35**

Conclusion:

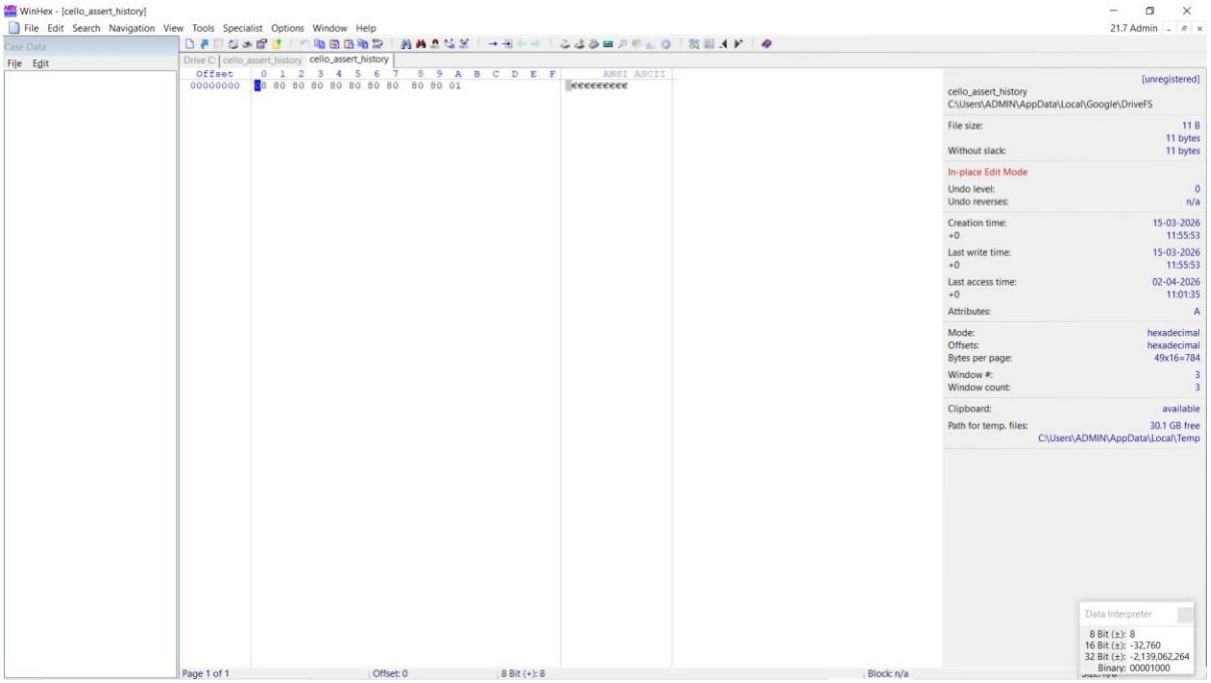
WinHex helps identify **last access time of Google Drive activity**

Key Points:

- Look for **Last Modified / Last Access Time**
- Analyze binary/hex values
- Convert timestamps if needed

Conclusion:

WinHex helps identify **when Google Drive was last accessed**, useful in investigations.



- WinHex showing registry or hex data

📄 Digital Forensic II – Slip 3 Solution

✅ Q1: What insights were gained by using Wireshark to generate a list of the top 10 websites visited by users in the network?

Steps:

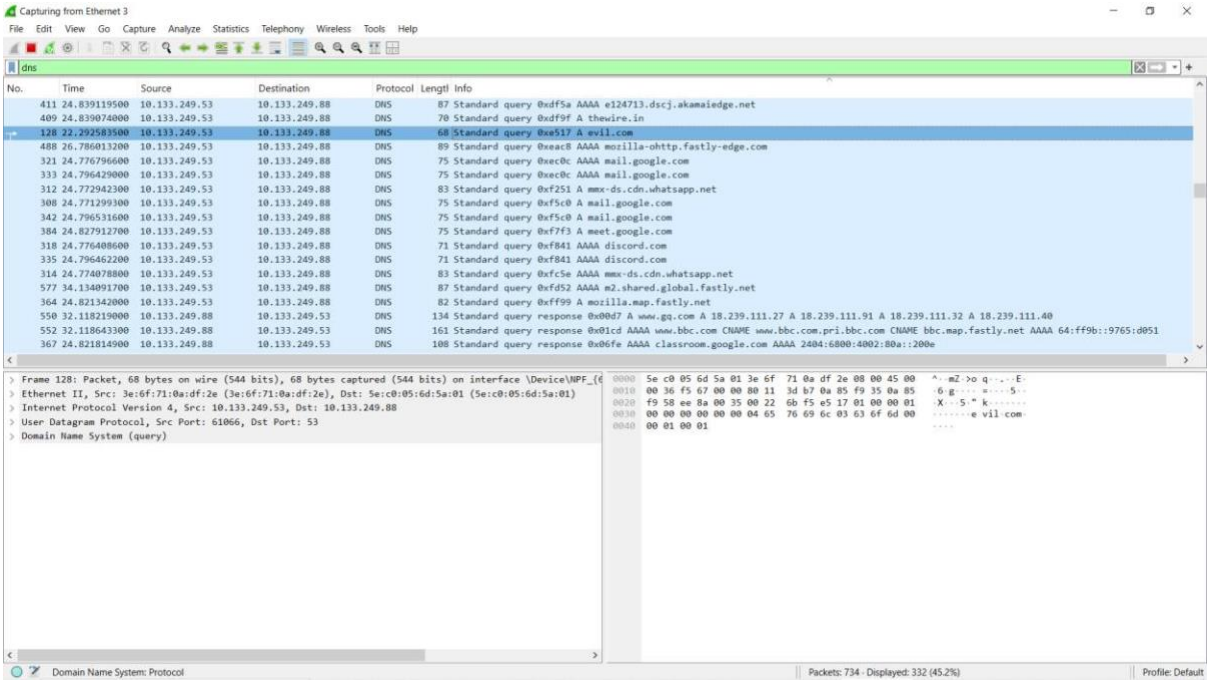
1. Open **Wireshark**
2. Select network interface 3. Start packet capture
4. Apply filter:
5. `http OR dns` 6. Go to:
    - **Statistics → HTTP → Requests**
    - OR **Statistics → Endpoints / Conversations**
7. Identify top visited domains

Insights Gained:

- List of **most visited websites**
- User browsing behavior
- Suspicious or unknown domains
- Frequency of visits

Conclusion:

Wireshark helps analyze **network traffic** and identify user activity.



- Wireshark showing HTTP/DNS traffic

✅ Q2: Solve the following questions

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a) Extract a list of all image, video, and audio files from a mobile device.

Using ADB

Steps:

1. Enable **USB Debugging** on mobile
2. Connect device to PC 3. Open terminal / CMD 4. Run command:

```
adb devices adb
shell
```

List Media Files:

```
ls /sdcard/DCIM/ ls
/sdcard/Music/ ls
/sdcard/Movies/
```

Extract Files:

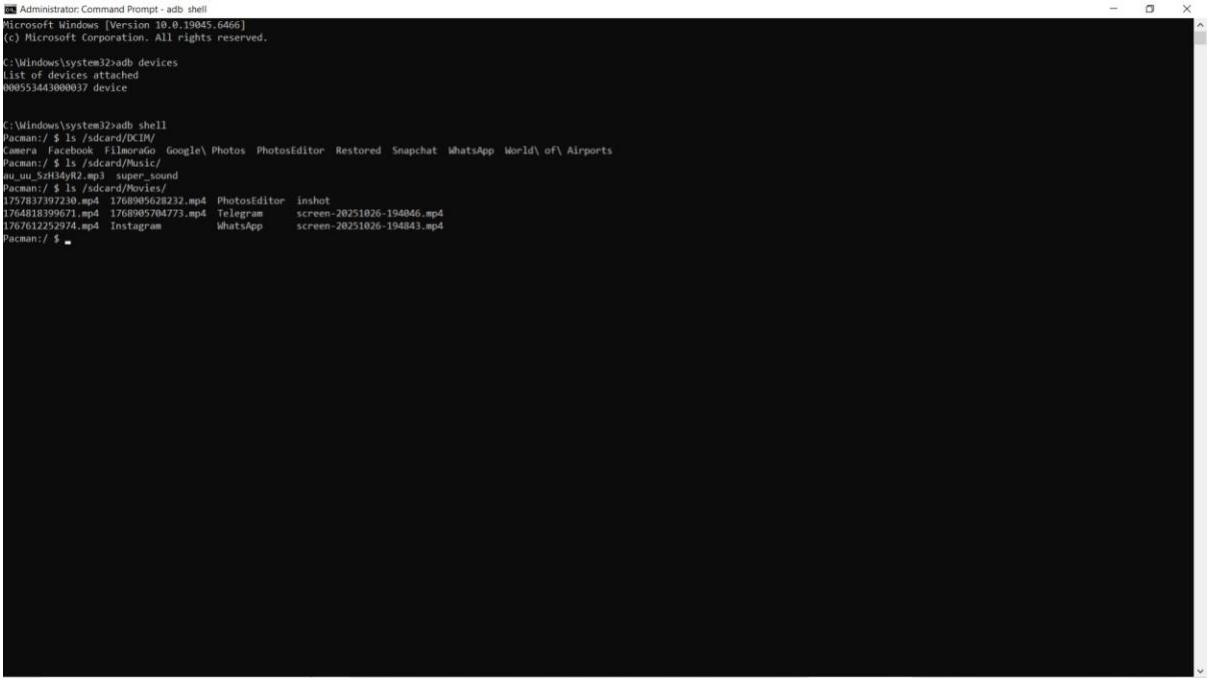
```
adb pull /sdcard/DCIM/ /local_folder/
adb pull /sdcard/Music/ /local_folder/
```

Result:

- Media files copied to system
- Can be analyzed further

Conclusion:

Media files can be extracted for **evidence analysis**



Screenshot of adb after getting shell access of mobile device.

---

b) How can you retrieve metadata (e.g., date created, location) for these files?

Steps

1. Create New Case

1. Open **Autopsy**
2. Click **Create New Case**
3. Enter case name and details

---

2. Add Data Source

4. Click **Add Data Source** 5.
Select:
   ☞ **Logical Files (IMPORTANT)**
6. Browse and select folder containing:
   - Images / Videos / Audio files

---

3. Configure Ingest Modules

7. Enable modules:
   - File Type Identification ○ EXIF Parser ○ Recent Activity
8. Click **Finish** to start analysis

---

4. Analyze Files

9. Go to:
   - **Data Sources → File Views → Images / Videos**
10. Select any media file

---

5. View Metadata

11. Click on file → check **Metadata panel (right side)**

Metadata includes:

- File name
- File size
- Creation date
- Modified date
- Camera/device info
- GPS location (if available)

---

6. Timeline Analysis

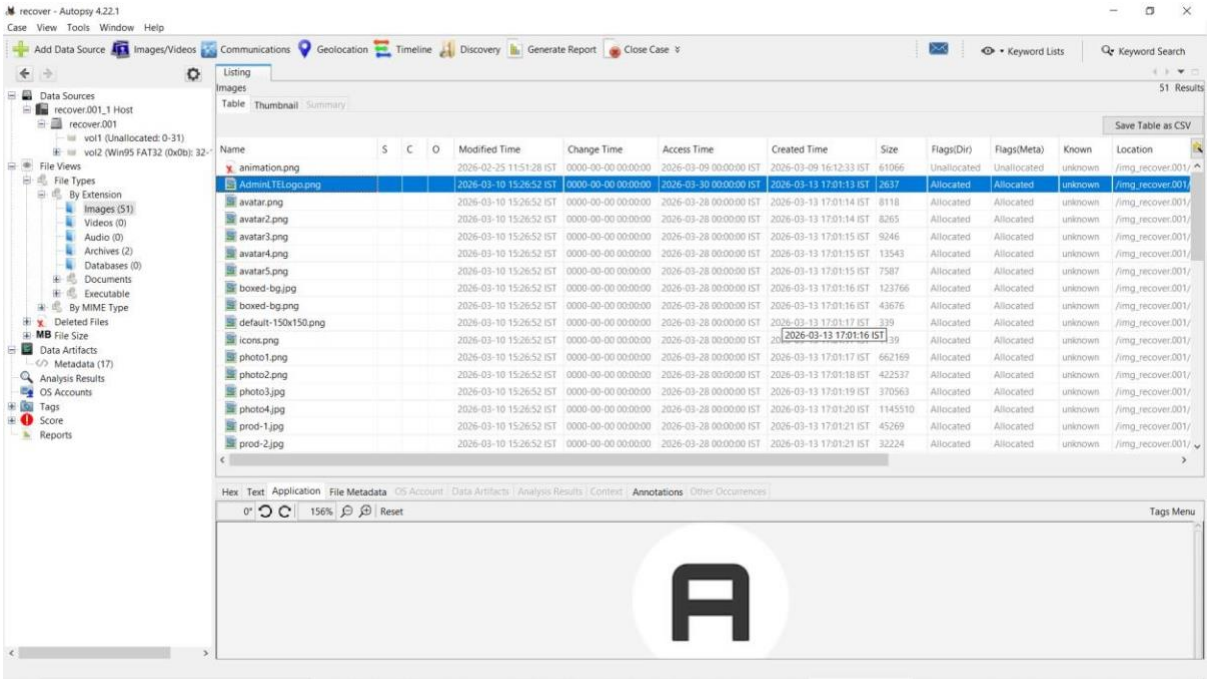
12. Go to **Timeline tab**
13. View activity based on timestamps

✅ Result:

Metadata of media files is successfully extracted and analyzed.

✅ Conclusion:

Autopsy helps retrieve important metadata like **date, time, and location**, useful in forensic investigations.



- Autopsy showing metadata panel of selected file

📄 Digital Forensic II – Slip 4 Solution

✅ Q1: Open an email and extract its header. Identify the following:

Steps:

1. Open email (Gmail/Outlook)
2. Click **"Show Original" / "View Header"**
3. Analyze header fields

Required Information:

- **a) Sender's Email Address:**
  → Found in **From** field
  b) Recipient's Email Address:
  → Found in **To** field
- **c) Date & Time of Sending:**
  → Found in **Date** field
- **d) Email Servers Used:**
  → Found in **Received** fields
  (Read **bottom → top** to trace path)

Conclusion:

Email headers provide **complete transmission details** of an email.

✅ Q2: Solve the following questions

---

a) How did you use Autopsy to analyze the Android image, and what options does it provide for file recovery?

✅ Tools Used:

- Autopsy
- ADB (for image acquisition)

✅ Step 1: Create Android Image

Method: Using ADB (Logical Image)

1. Enable **Developer Options** on Android device
2. Turn ON **USB Debugging**
3. Connect device to PC via USB 4. Open Command Prompt / Terminal 5. Check device connection: `adb devices`

6. Extract data:

```
adb pull /sdcard/ /local_folder/ (or whatever your path according to device)
```

👉 This creates a **logical image** of the device

Step 2: Load Image in Autopsy

1. Open **Autopsy**
2. Click **Create New Case**
3. Enter case details 4. Click **Add Data Source**

👉 Select:

- **Logical Files** (for ADB extracted data)
  OR

5. Browse and select image/folder 6. Enable ingest modules:
   - File Type Identification
   - Keyword Search
   - Recent Activity
7. Click **Finish** to start analysis

Step 3: Analyze Data

1. Navigate to:
   - **File Views**
   - **Data Sources**
2. Examine:
   - Images, videos, documents

Application data ○
Logs and databases

Step 4: File Recovery Options in Autopsy

Autopsy provides the following recovery features:

- **Deleted File Recovery** → Recovers deleted files from unallocated space
- **Keyword Search** → Finds specific data using keywords
- **Timeline Analysis** → Shows activity based on timestamps
- **File Type Filtering** → Filters files by type (.jpg, .mp4, etc.)
- **Hash Analysis** → Verifies file integrity

Result:

Android image is successfully analyzed and relevant files are recovered.

Conclusion:

Autopsy enables efficient analysis of Android images and supports recovery of deleted and existing data for forensic investigation.

📄 Digital Forensic II – Slip 6 Solution

✅ Q1: Solve the following

---

a) How did you use the tcpdump command-line program to examine network traffic?

Steps:

1. Open terminal (Linux/Kali) 2.
Run command:
3. `tcpdump -i eth0`
4. Capture packets from network interface 5. Use options:
   - `-c` → limit number of packets
   - `-w file.pcap` → save capture

Example Commands:

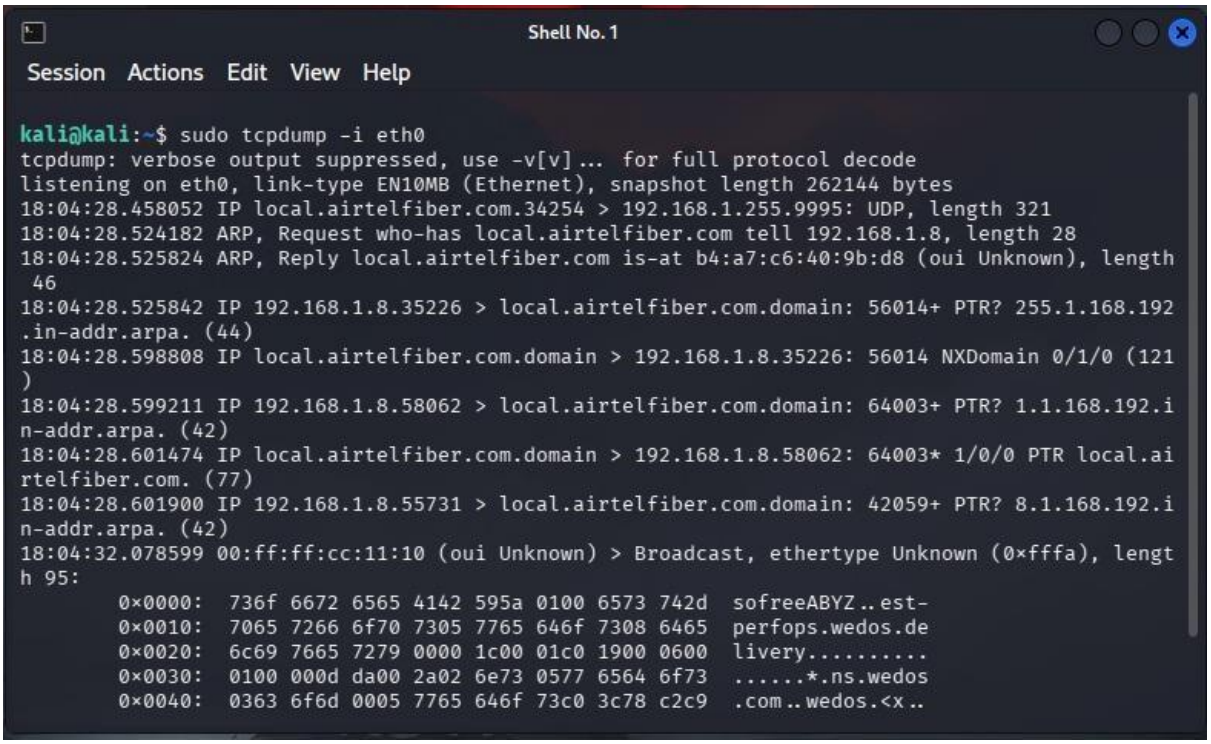
```
tcpdump -i eth0 -c 10 tcpdump -
i eth0 -w capture.pcap
```

Analysis:

- View source & destination IP
- Identify protocols (HTTP, DNS, TCP)
- Detect suspicious traffic

Conclusion:

tcpdump helps in **capturing and analyzing network packets**



✅ Q2: Solve the following

---

a) Document the extracted data, including time of extraction and method used.

Steps:

1. Record **date & time of extraction**
2. Mention **tool used** (Wireshark, Autopsy, etc.)
3. Describe **method used**
4. Save extracted data securely

---

b) What key details should be included in a forensic data report?

Include:

- Case details
- Investigator name
- Tools used
- Steps followed
- Evidence collected
- Timestamps
- Conclusion

Conclusion:

Proper documentation ensures **evidence integrity and legal validity**`

📄 Digital Forensic II – Slip 7 Solution

✅ Q1: Solve the following questions

a) What information can be obtained by examining files associated with VMs, especially log files, in the context of investigating a crime or incident?

VM files include:

- `.vmdk` → Virtual disk
- `.vmx` → Configuration

- `.log` → Activity logs

Information obtained:

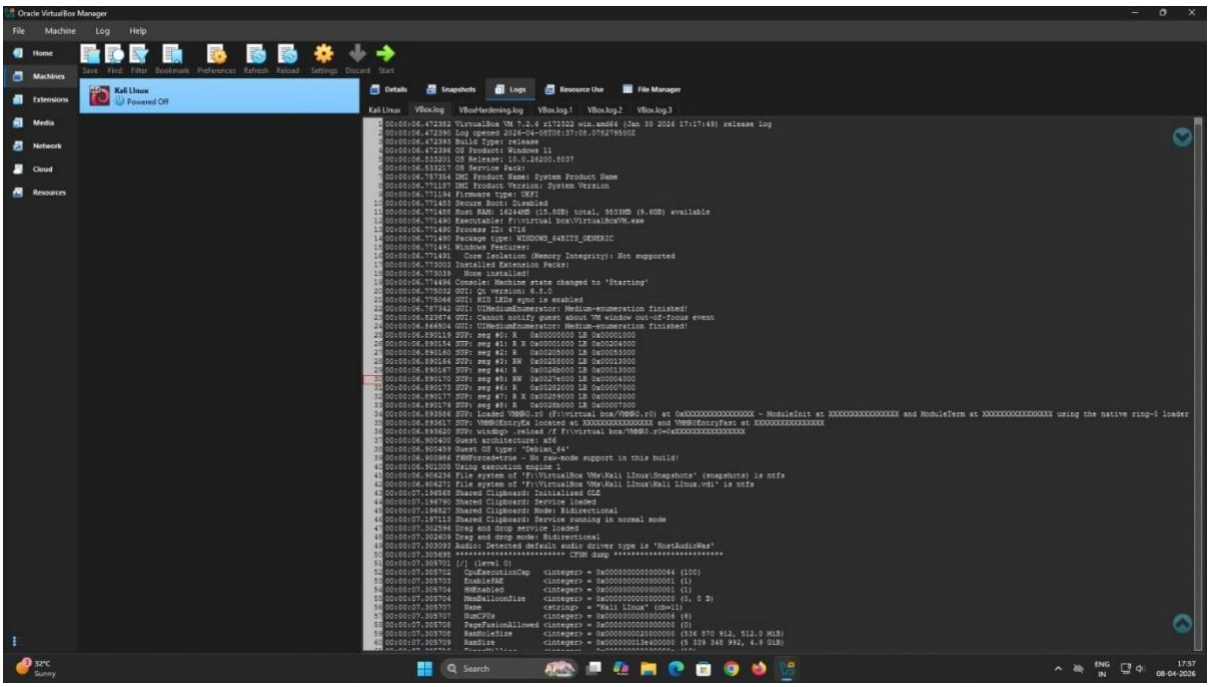
- VM **start/stop time**
- User **login activity**
- Installed software
- Network activity
- Errors or crashes

Log file analysis:

- Check timestamps
- Identify unusual activity
- Detect unauthorized access

Conclusion:

VM files help reconstruct **user actions and system events**



- VM log file (optional)

✅ Q2: Solve the following

---

a) Identify the SPF (Sender Policy Framework) record in an email header. What does it tell you about the legitimacy of the sender?What it is:

- Email authentication method

Where found:

- In email header

What it tells:

- Whether sender IP is **authorized**
- Helps detect spoofing

👉 Example:

- **SPF = pass** → Legit
- **SPF = fail** → Suspicious

---

b) Find the DKIM (Domain Keys Identified Mail) signature in the email header. What domain signed the email, and was it verified successfully?

What it is:

- Digital signature of email

What it shows:

- Which domain signed the email □ Whether message is **modified or not**

👉 Example:

- **DKIM = pass** → Email not altered
- **DKIM = fail** → Possible tampering

Conclusion:

SPF and DKIM help verify **email authenticity and integrity**

📄 Digital Forensic II – Slip 8 Solution

✅ Q1: Solve the following questions

a) Extracting device information of your Android device using ADB. Create a report of your findings.

Steps:

1. Enable **Developer Options** on Android
2. Turn ON **USB Debugging**
3. Connect phone to PC via USB 4. Open terminal / command prompt 5. Run commands:

```
adb devices adb
shell getprop
```

Important Information Extracted:

- **Manufacturer:** `ro.product.manufacturer`
- **Model:** `ro.product.model`
- **Android Version:** `ro.build.version.release`
- **Device ID:** shown in `adb devices`

Conclusion:

ADB is used to collect **basic device details for forensic analysis**

✅ Q2: Which tool or method did you use to create an image of your phone data, and why?

Tools / Methods:

- **ADB (Android Debug Bridge)**
- **Mobile forensic tools** (Autopsy, Cellebrite, etc.)

Steps using ADB:

1. Connect device with USB debugging ON
2. Run command: `adb pull /sdcard/ /local_folder/`
3. Copy all data from device to system

Why this method is used:

- Safe data acquisition
- Maintains data integrity
- Easy to perform

Conclusion:

Imaging creates a **copy of device data for investigation without altering original data**

📄 Digital Forensic II – Slip 9 Solution

✅ Q1: Solve the following questions

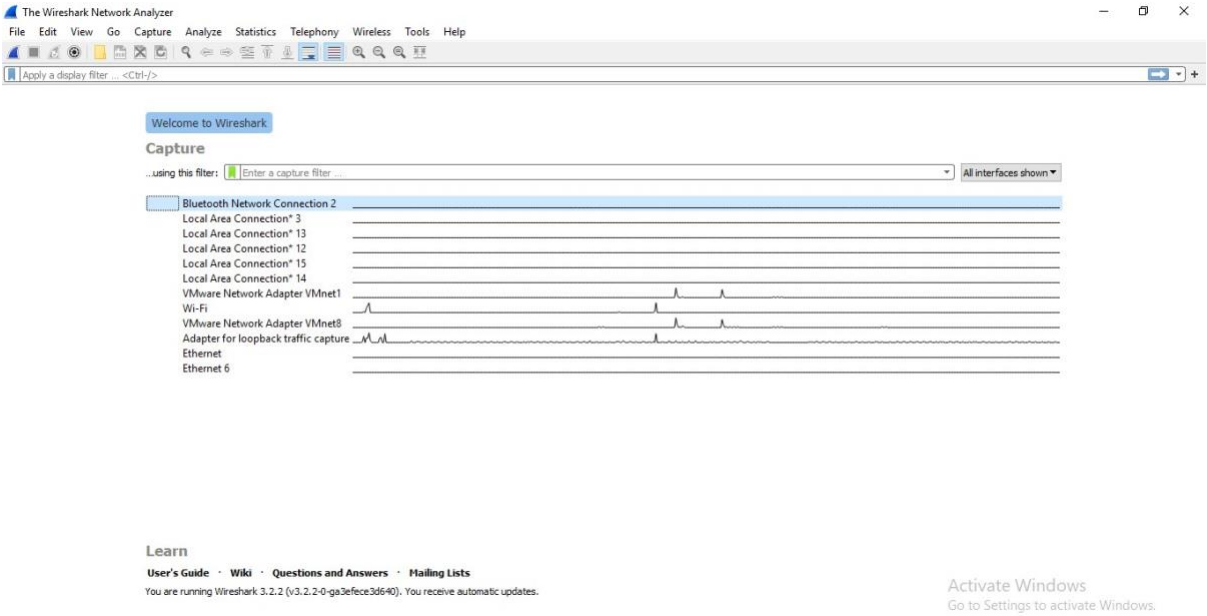
---

a) Using Wireshark, how do you generate a list of the top 10 websites accessed from a VM?

Steps:

1. Open **Wireshark (you can open wireshark directly in vm or inside your host machine only select your vm adapter as given inside screenshot)**
2. Select network interface 3. Start packet capture
4. Apply filter:
5. `http OR dns` 6. Go to:
    - **Statistics → HTTP → Requests**
    - OR **Statistics → Endpoints / Conversations**
7. List top domains based on traffic

Conclusion:



Wireshark helps identify **most visited websites and user activity**

---

b) What indicators can be used to detect data exfiltration, unauthorized access, or malware communication within a network?

Common indicators:

- Unusual or unknown IP addresses
- Repeated failed connections
- Large data transfers (data exfiltration)
- Communication with external servers
- Suspicious protocols or ports
- Malware traffic patterns

Conclusion:

These indicators help detect **unauthorized access and malware activity**

✅ Q2: Solve the following questions

---

a) How can you extract and analyze call logs from the mobile device image?

Steps:

1. Enable **USB Debugging** on Android device
2. Connect device to PC
3. Open Command Prompt / Terminal 4. Check connection: `adb devices`

Access Call Log Database:

5. Open shell: `adb shell`

6. Navigate to path:

`/data/data/com.android.providers.contacts/databases/`

Extract Call Log File:

7. Locate database file:

- `contacts2.db` (contains call logs)

8. Pull file:

```
adb pull /data/data/com.android.providers.contacts/databases/contacts2.db
```

Data extracted:

- Phone numbers
- Call duration
- Date & time
- Incoming/outgoing/missed

b) How to recover deleted data can be recovered from the mobile image

Steps:

---

1. Load image in **Autopsy (you can use adb to create android image as given in above questions)**
2. Use:
   - Deleted file recovery
   - Keyword search
3. Analyze unallocated space

Conclusion:

Deleted data can be recovered and used as **digital evidence**

📄 Digital Forensic II – Slip 10 Solution

✅ Q1: Solve the following questions

---

a) How can you use ADB (Android Debug Bridge) to retrieve basic device information such as manufacturer, model, and OS version?

Steps:

1. Enable **USB Debugging** on Android
2. Connect device to PC 3. Open terminal / CMD 4. Run

command: `adb shell getprop`

Information Retrieved:

- **Manufacturer:** `ro.product.manufacturer`
- **Model:** `ro.product.model`
- **OS Version:** `ro.build.version.release`

Conclusion:

ADB helps extract **basic system details for forensic analysis**

---

b) What ADB command retrieves a list of installed applications on an Android device? `adb shell pm list packages`

☞ Shows all installed apps on device

Conclusion:

Helps identify **installed and suspicious applications**

✅ Q2: Solve the following questions

---

a) Look for the DMARC (Domain-based Message Authentication, Reporting & Conformance) record. What policy is set for handling failed authentication?

What is DMARC?

- Email authentication policy

What it shows:

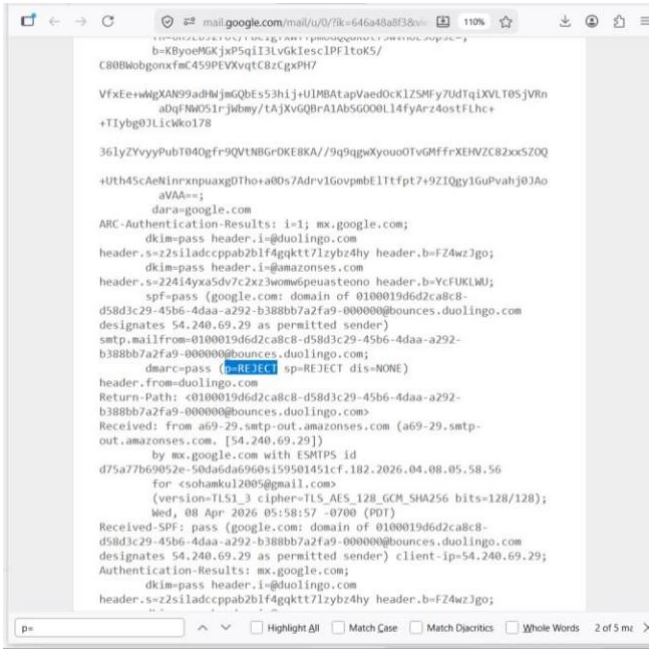
- Action when SPF/DKIM fails

☞ Policies:

- **none** → No action
- **quarantine** → Move to spam
- **reject** → Block email

Conclusion:

DMARC helps prevent **email spoofing and phishing**



Header showing policy that is p=REJECT

---

b) In what time zone does the sender's email server reside? What header field provides this information ?

Where to find:

- **Date field in email header**

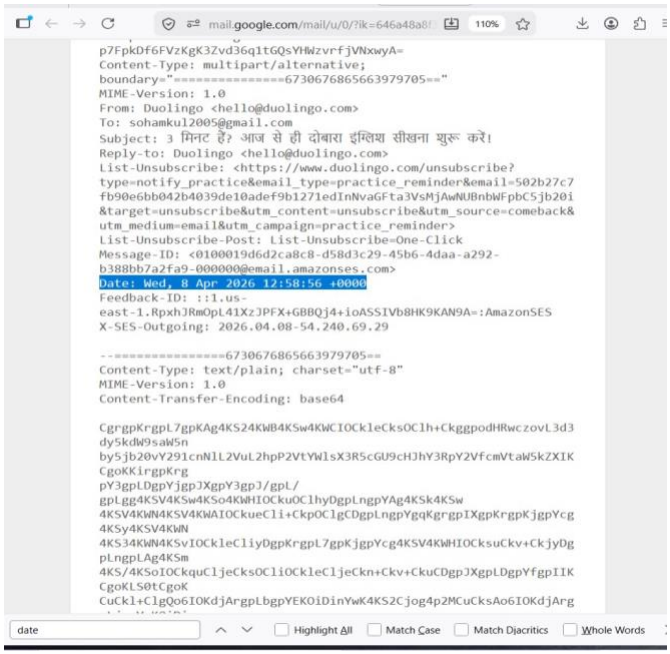
☞ Example:

```
Date: Mon, 10 Feb 2025 10:30:00 +0530
```

- `+0530` → Time zone (India)

Conclusion:

Time zone helps identify **sender location**



📄 Digital Forensic II – Slip 12 Solution

✅ Q1: Solve the following questions

a) How can you extract and analyze metadata from images and videos found in the mobile device image

---

Aim:

To extract and analyze metadata from images and videos in a mobile device image.

Tools Used:

- Autopsy
- ExifTool

Steps (Using Autopsy):

1. Open **Autopsy**

2. Create **New Case** 3. Click **Add Data Source** 4. Select:
    - Logical Files / Disk Image
6. Load mobile image
7. Enable ingest modules:
    - File Type Identification ○ EXIF Parser

Analysis:

8. Navigate to:
    - File Views → Images / Videos
9. Select a media file
10. View metadata in **right panel**

Metadata Extracted:

- File name
- File size
- Creation date
- Modified date
- Device/camera info
- GPS location (if available)

Conclusion:

Metadata helps identify **when, where, and how the file was created**

✅ Q2: Solve the following questions

---

a) What permissions are required to access system-level information using ADB?

Requirements:

- Enable **Developer Options**
- Enable **USB Debugging**

- Grant **device authorization** (Allow USB debugging prompt) □ Root access (for system-level data)

Conclusion:

ADB requires proper permissions to access **device and system data**

---

b) What security risks are associated with enabling ADB on a mobile device?

Risks:

- Unauthorized access to device
- Data theft
- Malware installation
- Bypass of security controls
- Exposure of sensitive data

Conclusion:

ADB should be used carefully as it can expose **serious security vulnerabilities**

📄 Digital Forensic II – Slip 17 Solution

✅ Q1: Solve the following questions

---

a) Using (MXToolbox, Email Header Analyzer) analyze email header,

Steps:

1. Open email → click **"Show Original"**
2. Copy full **email header** 3. Go to:
    - MXToolbox (Header Analyzer)

4. Paste header and analyze

Information obtained:

- Sender IP address
- Mail servers involved
- Authentication results

---

b) Find out SPF, DKIM, and DMARC records, and how do they help in email authentication

---

1. SPF (Sender Policy Framework)

- Verifies sender IP
- **Pass → Legit**
- **Fail → Spoofing possible**

---

2. DKIM (DomainKeys Identified Mail)

- Digital signature
- Ensures email not modified

---

3. DMARC

- Policy for failed authentication

☞ Policies:

- **none** → No action
- **quarantine** → Spam folder
- **reject** → Block email

Conclusion:

These protocols ensure **email authenticity and security**

✅ Q2: Solve the following questions

---

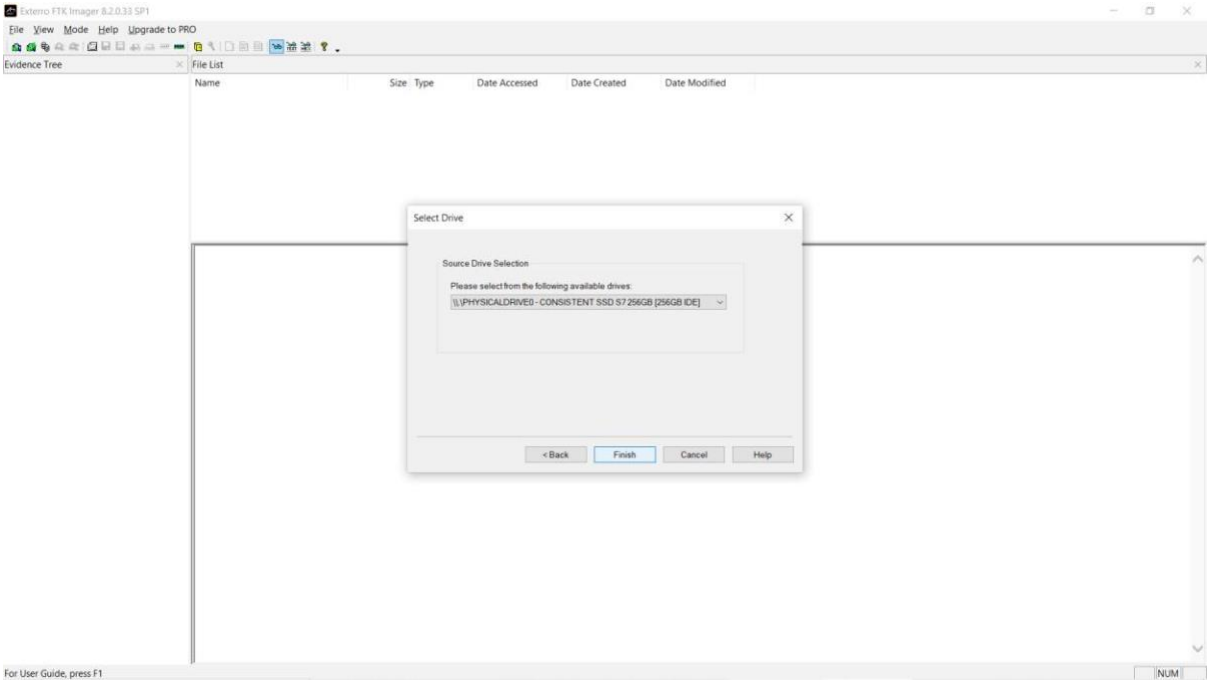
a) How can you acquire a forensic image of a host machine running multiple VMs?

Steps:

1. Use tool:
    - **FTK Imager**
2. Select physical drive
3. Create image (E01 / RAW format)
4. It will automatically Include VM files:
    - `.vmdk`, `.vmx`, `.log`

Conclusion:

Imaging captures **complete system data including VMs**



- FTK Imager acquisition of host machine selct your main drive where vm is installed and start imaging

---

b) What types of network logs (e.g., firewall, IDS/IPS, system logs) can be used to trace activities between the host and VMs?

Types of logs:

- Firewall logs IDS/IPS logs

- System logs
- Network traffic logs (Wireshark)

What they show:

- Communication between host & VM
- Suspicious connections
- Data transfer

Conclusion:

Logs help trace **interaction and possible attacks**

📄 Digital Forensic II – Slip 19 Solution

✅ Q1: Solve the following questions

---

a) What types of network logs (e.g., firewall, IDS/IPS, system logs) can be used to trace activities between the host and VMs? Types of logs:

- **Firewall logs** → Allowed/blocked connections
- **IDS/IPS logs** → Intrusion detection alerts
- **System logs** → OS-level activity
- **Network logs (Wireshark/tcpdump)** → Packet-level data

What they reveal:

- Communication between **host and VM**
- Suspicious IP addresses
- Unauthorized access attempts ▫ Data transfer details

Conclusion:

Network logs help trace **interaction and possible attacks**

---

b) How can you use system logs to determine when a VM was created, modified, or deleted?

Steps:

1. Access system/hypervisor logs
2. Check for events: ○ VM **creation** ○ VM **start/stop**
   ○ VM **modification/deletion**
3. Analyze timestamps

Conclusion:

System logs help track **complete VM lifecycle**

✅ Q2: Solve the following questions

---

a) What tools can be used to recover deleted SMS messages from an Android device?

Tools:

- **Autopsy**
- **ADB**

Steps:

1. Acquire mobile image (using adb)
2. Load into forensic tool
3. Locate SMS database (`.db file`)
4. Recover deleted messages

---

b) How can you analyze the recovered SMS messages to verify their integrity?

Methods:

- Check timestamps
- Verify sender/receiver details
- Use **hash values**
- Cross-check with logs

Conclusion:

Recovered SMS can be used as **valid forensic evidence**

📄 Digital Forensic II – Slip 20 Solution

✅ Q1: Solve the following questions

a) How can you identify the originating IP address of an email, and what tool can be used to perform a reverse lookup?

Steps:

1. Open email → click **"Show Original / View Header"**
2. Find **Received fields**
3. Read **bottom → top**
4. Identify first **public IP address**

Reverse Lookup Tool:

- **MXToolbox** / IP lookup tools

👉 Paste IP → get:

- Location
- ISP

✅ Q2 : Solve the following questions

a) How can you identify the top 10 websites visited by a VM user using Wireshark?

Steps:

1. Open **Wireshark**
2. Start capture 3.
Apply filter:

```
http OR dns
```

4. Go to:

- **Statistics → HTTP Requests**
- OR **Endpoints / Conversations**

5. List top domains

---

b) What forensic indicators can be found in network packets that reveal communication between a VM and external servers?

- Unknown/external IP addresses
- DNS queries to suspicious domains
- Repeated connections
- Unusual ports/protocols
- Large data transfer (exfiltration)
- Encrypted traffic to unknown servers

✅ Conclusion

Email headers reveal **sender IP**, and network packets show **communication & suspicious activity**

📄 Digital Forensic II – Slip 24 Solution

✅ Q1. Solve the following questions

a) How do you acquire a forensic image of a host machine running a VM using FTK Imager Lite?

Steps:

1. Open **FTK Imager Lite**
2. Click **File → Create Disk Image** 3. Select **Physical Drive** (entire SSD/HDD) 4. Choose image format: ○ E01 / RAW
5. Select destination path
6. Start imaging process

Important Point:

- Imaging the **host machine** automatically includes:
  - VM files (`.vmdk`, `.vmx`, `.log`)

Conclusion:

A forensic image of the host machine captures **complete data including virtual machines**

---

b) What types of data should be collected for forensic analysis?

Data to collect:

- **System logs** → OS activity
- **Network logs** → communication details
- **VM disk images** → `.vmdk`, `.vdi` **VM logs** → `.log files`
- **Registry files** → user/system activity
- **Application logs** → software usage

Conclusion:

Collecting multiple data sources helps perform **complete forensic investigation**

✅ Q2. Solve the following questions

---

a) How can WinHex be used to analyze the Windows registry for Google Drive activity?

Steps:

1. Open **WinHex**
2. Go to **Tools → Open Disk**
3. Select system drive
4. Navigate to registry location:

```
C:\Users\<User>\NTUSER.DAT
```

5. Open registry file
6. Search for:

```
Google / Drive
```

7. Analyze entries for:
   - Access time
   - Usage details

Conclusion:

WinHex helps analyze registry to find **Google Drive activity and timestamps**

---

b) What specific registry keys store information about the last access time of Google Drive?

Registry Keys:

```
HKEY_CURRENT_USER\Software\Google\Drive
HKEY_CURRENT_USER\Software\Google\DriveFS
```

Data Stored:

- Last access time
- Sync activity
- File usage details

Conclusion:

These registry keys store **Google Drive usage and access information**